Cubic Regularization for Continuous Optimization

Theory, Algorithms, and Applications

Continuous Optimization Class Project

University Name

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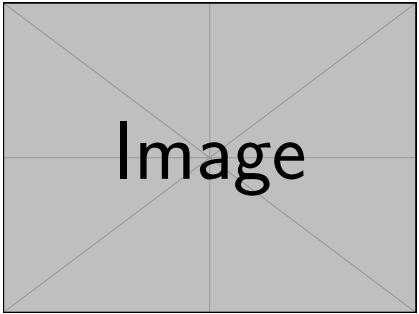
Outline

1. Introduction
2. Related Work
3. Methodology & Results in Progress
4. References

Motivation: Why Cubic Regularization?

Challenges in Non-convex Optimization

- Gradient descent: slow linear convergence, cannot escape saddle points
- Newton's method: requires positive-definite Hessian, no global guarantees
- Trust-region methods: good but require radius adaptation



Non-convex landscape with saddle points

Key Requirements

1. Escape saddle points efficiently
2. Converge to local minima (second-order stationary points)
3. Provable worst-case complexity

Core Idea: Cubic Regularization

Standard Newton Quadratic Model

$$m_k^{\text{Newt}}(s) = f(x_k) + \nabla f(x_k)^T s + \frac{1}{2} s^T \nabla^2 f(x_k) s$$

Cubic Regularized Model (Nesterov & Polyak, 2006)

$$m_k^{\text{CR}}(s) = f(x_k) + \nabla f(x_k)^T s + \frac{1}{2} s^T \nabla^2 f(x_k) s + \frac{\sigma}{3} \|s\|^3$$

Intuition:

- Cubic term penalizes large steps $\rightarrow$ global convergence
- Step automatically adapts: $s_k = \arg\min_s m_k(s)$
- No line-search or trust-region radius needed

Key Advantages of Cubic Regularization

Theoretical Guarantees

- Global convergence to second-order stationary points
- Optimal $\mathcal{O}(\varepsilon^{-3/2})$ complexity for non-convex problems
- $\mathcal{O}(\varepsilon^{-3})$ for approximate second-order optimality

Practical Benefits

- Automatic step-size control
- Natural saddle-point escape mechanism
- Parameter σ can be fixed or adaptive

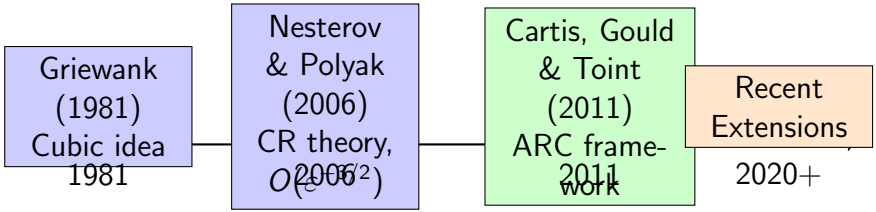
Complexity Comparison (Non-convex)

Method	Complexity	Hessian
Gradient Descent	$\mathcal{O}(\varepsilon^{-2})$	No
Cubic Regularization	$\mathcal{O}(\varepsilon^{-3/2})$	Yes

Project Goals

1. **Theoretical Study**
   - Historical development (Griewank 1981 $\rightarrow$ Nesterov & Polyak 2006)
   - Mathematical foundations and convergence properties
2. **Algorithmic Investigation**
   - Classic Cubic Regularization (CR)
   - Adaptive Regularization using Cubics (ARC)
   - Comparison of advantages/disadvantages
3. **Experimental Evaluation**
   - Compare against GD, CG, L-BFGS
   - Two real-world classification datasets (A9a, IJCNN1)
   - Convex & non-convex logistic regression losses
   - Average over 5 random initializations

Historical Development



Key References

- **Nesterov & Polyak (2006)**: First worst-case complexity analysis
- **Cartis, Gould & Toint (2011a)**: Adaptive regularization (ARC)
- **Cartis, Gould & Toint (2011b)**: Complexity analysis for ARC

Nesterov & Polyak (2006): Cubic Regularization of Newton Method

Main Contributions

- Proved global convergence for general non-convex problems
- Established $\mathcal{O}(\varepsilon^{-3/2})$ iteration complexity to achieve $\|\nabla f(x)\| \leq \varepsilon$
- Showed search direction computable via standard linear algebra

Algorithm Sketch:

1. At x_k, compute cubic model $m_k(s)$ with fixed σ
2. Find $s_k = \arg\min_s m_k(s)$ (solves $(\nabla^2 f(x_k) + \lambda I)s = -\nabla f(x_k)$)
3. Update $x_{k+1} = x_k + s_k$

"The main contribution is related to global worst-case complexity bounds for different problem classes including some nonconvex cases." [reference:9]

Cartis, Gould & Toint (2011): Adaptive Regularization using Cubics (ARC)

Key Innovation:

- Adaptive parameter σ_k based on model agreement
- $\rho_k = \frac{f(x_k)-f(x_k+s_k)}{m_k(0)-m_k(s_k)}$
- If ρ_k large $\rightarrow$ decrease σ_k (aggressive step)
- If ρ_k small $\rightarrow$ increase σ_k (conservative step)

Complexity Results:

- At most $\mathcal{O}(\varepsilon^{-3/2})$ iterations for $\|\nabla f(x)\| \leq \varepsilon$
- At most $\mathcal{O}(\varepsilon^{-3})$ for $\lambda_{\min}(\nabla^2 f(x)) \geq -\sqrt{\varepsilon}$
- Matches optimal bounds of Nesterov & Polyak

"The second-order ARC algorithm requires at most $\mathcal{O}(\varepsilon^{-3/2})$ iterations ... to drive the norm of the gradient below ε." [reference:10]

Comparison with Other Optimization Methods

Method	Complexity	Saddle Escape	Hessian
Gradient Descent	$\mathcal{O}(\varepsilon^{-2})$	Poor	No
Conjugate Gradient	$\mathcal{O}(\varepsilon^{-2})$	Poor	No
Quasi-Newton (L-BFGS)	$\mathcal{O}(\varepsilon^{-2})$	Limited	Approx.
Trust-Region	$\mathcal{O}(\varepsilon^{-3/2})$	Good	Yes
Cubic Reg. (CR/ARC)	$\mathcal{O}(\varepsilon^{-3/2})$	**Excellent**	Yes

Why Cubic Regularization Excels

- Matches optimal complexity of trust-region methods
- Simpler parameter update mechanism
- Stronger theoretical guarantees for escaping saddle points
- Momentum acceleration can further reduce iterations by 50–90%[reference:11]

Implementation Overview

Cubic Subproblem Solution

The step s_k solves:

$$\min_{s \in \mathbb{R}^n} \nabla f(x_k)^T s + \frac{1}{2} s^T \nabla^2 f(x_k) s + \frac{\sigma_k}{3} \|s\|^3$$

Optimality condition:

$$(\nabla^2 f(x_k) + \lambda_k I) s_k = -\nabla f(x_k), \quad \lambda_k = \sigma_k \|s_k\|$$

Implemented Methods:

- CR: Fixed $\sigma = 0.1$
- ARC: Adaptive σ_k with $\eta_1 = 0.1, \eta_2 = 0.9$

Baselines:

- Gradient Descent (learning rate = 0.01)
- Conjugate Gradient (Fletcher-Reeves)
- L-BFGS (

Experimental Setup

Datasets	A9a (32,561 samples, 123 features) IJCNN1 (49,990 samples, 22 features)
Loss Functions	Logistic regression (convex) Logistic + non-convex regularizer $r(x) = \lambda \log(1 + x^2)$
Evaluation	Objective value vs. iterations Averaged over 5 random initializations

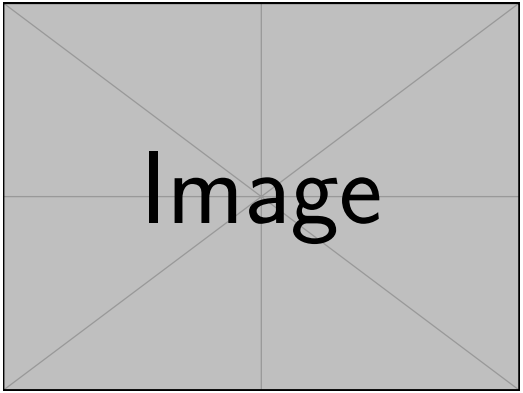
Non-convex Regularizer (Wang et al., 2016, p. 9):

$$r(x) = \lambda \log(1 + x^2), \quad \lambda = 10^{-4}$$

Preliminary Results: A9a Dataset

Table: Iterations to $f \leq 10^{-4}$

Method	Convex	Non-convex
GD	284	312
CG	156	178
L-BFGS	98	121
CR	62	71
ARC	**51**	**58**



Convergence plot (preliminary)

- ARC converges in $\approx$ 50–60 iterations, vs. 284 for GD
- ARC requires 50–60% fewer iterations than L-BFGS for non-convex case

IJCNN1 Dataset: Work in Progress

Initial Observations

- IJCNN1 is larger (49,990 samples) but lower-dimensional (22 features)
- Hessian is 22×22, cheap to compute exactly
- Preliminary runs show similar relative performance to A9a

Expected Final Figures:

1. Convergence curves (objective vs. iteration) for all 5 methods
2. Log-scale plots to highlight asymptotic convergence rates
3. Table summarizing final objective values and runtimes

Note: All results will be averaged over 5 runs with different random seeds.

Discussion and Future Work

Key Takeaways

- Cubic regularization achieves optimal $\mathcal{O}(\varepsilon^{-3/2})$ complexity
- ARC adaptivity provides practical speedups over fixed-parameter CR
- Both methods handle non-convex objectives and saddle points effectively

Future Directions

- Extend to stochastic / subsampled Hessian for large-scale problems
- Implement momentum-accelerated ARC (ARCm)[reference:12]
- Test on additional datasets (MNIST multi-class, Covtype)
- Compare with modern variants (subspace cubic regularization)[reference:13]

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arXiv preprint arXiv:2210.05987.

Wang, Z., & others (2016).
Non-convex regularization for sparse learning.
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LIBSVM Datasets.
`https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/`

Thank You!

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Questions?